Introduction to Data Science and Machine Learning

Research Assignment

Monday, June 15, 2026

Introduction

Data Science and Machine Learning are important fields that help solve real-world problems. They are used in healthcare, banking, self-driving cars, and language technology. Data Science focuses on collecting and analyzing data, while Machine Learning helps computers learn from data and make predictions. This assignment explains the basic ideas of these fields, the steps in a data science project, and how machine learning models are trained and tested. The information comes from trusted academic and technical sources.

1. Defining Data Science and Machine Learning

1.1 Data Science

Data Science is the study of collecting, organizing, and analyzing data to find useful information. It uses mathematics, statistics, computer science, and technology to help people make better decisions. Data scientists work with different types of data and make sure the data is useful and accurate.

1.2 Machine Learning

Machine Learning (ML) is a part of Artificial Intelligence (AI). It allows computers to learn from data and improve their performance without being programmed for every task. ML finds patterns in data and uses them to make predictions or classify information. It is widely used in applications like recommendation systems, image recognition, and fraud detection.

1.3 The Relationship Between Them

Data Science and Machine Learning are connected but different. Data Science is the larger field that includes collecting, cleaning, analyzing, and presenting data. Machine Learning is one of the tools used in Data Science to build models and make predictions. In simple terms, Data Science works with data, while Machine Learning helps computers learn from that data.

1.4 Real-Life Example: Healthcare

A hospital wants to know which patients may return within 30 days after leaving the hospital.

Data Science is used to collect and prepare patient information, such as:

- Age
- Medical history
- Number of medicines
- Previous hospital visits

The data is cleaned and organized before analysis.

Machine Learning is then used to study old patient records and learn patterns. The model predicts which new patients are at high risk of returning to the hospital. Doctors can use these predictions to provide extra care and reduce readmissions.

Simple Example:
If many older patients with several health problems return to the hospital, the ML model learns this pattern and warns doctors when a similar patient arrives.

2. The Data Science Lifecycle

The Data Science Lifecycle is a series of steps used to solve problems with data. Each step is important for building a successful project.

2.1 Main Stages

Stage	What Happens
1. Problem Definition	Understand the problem and set goals.
2. Data Collection	Gather data from different sources.
3. Data Cleaning	Remove errors and organize the data.
4. Data Analysis	Study the data and find patterns.
5. Model Building	Use Machine Learning to create a model.
6. Model Evaluation	Check how accurate the model is.
7. Deployment	Put the model into real-world use and monitor it.

Simple Example: Predicting Student Grades

1. Define the goal: Predict final exam scores.
2. Collect student data.
3. Clean incorrect or missing information.
4. Find important factors like attendance and homework.
5. Train an ML model.
6. Test the model's accuracy.
7. Use the model to help teachers identify students who need support.

2.2 Where Machine Learning Fits

Machine Learning is mainly used during the **Model Building** stage. After the data is collected and cleaned, ML learns patterns from the data and makes predictions.

It is also used during:

- **Evaluation:** Checking if the model works well.
- **Monitoring:** Updating the model when new data becomes available.

Simple Example:
A weather app learns from past weather data to predict tomorrow's temperature. As new weather data arrives, the model can be improved.

3. Supervised vs. Unsupervised Learning

Machine Learning has two main types: Supervised Learning and Unsupervised Learning.

3.1 Supervised Learning

Supervised Learning uses **labeled data**, meaning the correct answer is already known. The model learns from these examples and predicts future results.

Common Uses:

- Spam email detection
- Fraud detection
- House price prediction
- Disease diagnosis

Simple Example:
A bank has thousands of transactions labeled as "Fraud" or "Not Fraud." The ML model learns these patterns and checks new transactions for possible fraud.

Another example:
A teacher gives a model past student scores and final grades. The model learns the relationship and predicts future student grades.

3.2 Unsupervised Learning

Unsupervised Learning uses **unlabeled data**, meaning there are no correct answers. The model looks for hidden patterns and groups similar data together.

Common Uses:

- Customer grouping
- Product recommendations
- Market analysis
- Finding unusual behavior

Simple Example:
An online shopping company studies customer buying habits. Without any labels, the ML model groups customers into categories such as:

- Budget shoppers
- Luxury buyers
- Seasonal customers

The company can then send different advertisements to each group.

Another example:
A music app groups users with similar music tastes and recommends new songs they may enjoy.

Difference Between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data.	Uses unlabeled data.
Knows the correct answers.	Finds hidden patterns on its own.
Used for prediction and classification.	Used for grouping and discovering patterns.
Example: Email spam detection.	Example: Customer segmentation.

Easy Way to Remember:

- **Supervised Learning:** The model learns with a teacher because the answers are provided.
- **Unsupervised Learning:** The model learns by itself because there are no answers given.

3.3 Supervised vs. Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data.	Uses unlabeled data.
Predicts outcomes.	Finds hidden patterns.
Needs known answers.	Does not need answers.
Example: Spam email detection.	Example: Customer grouping.

Easy to remember:

- **Supervised = Learn with answers.**
- **Unsupervised = Learn without answers.**

4. Overfitting: Causes and Prevention

4.1 What is Overfitting?

Overfitting happens when a Machine Learning model learns the training data too well, including mistakes and random details. As a result, it performs poorly on new data.

Example:
A student memorizes old exam questions instead of understanding the subject. If the exam changes, the student may fail.

Underfitting is the opposite. The model is too simple and cannot learn enough from the data.

4.2 Causes of Overfitting

Common causes are:

- Model is too complex.
- Not enough training data.
- Poor-quality or unnecessary data.
- Training for too long.
- No methods to control model complexity.

4.3 How to Prevent Overfitting

Some common solutions are:

- **Regularization:** Keep the model simple.
- **Cross-validation:** Test the model on different parts of the data.

- **Early stopping:** Stop training before the model memorizes the data.
- **Pruning:** Remove unnecessary parts of the model.
- **More training data:** Give the model more examples to learn from.

Simple Example:
A teacher gives students many different practice questions instead of making them memorize one test.

5. Training Data and Test Data

5.1 What is Train-Test Split?

Before training a Machine Learning model, the data is divided into two parts:

- **Training Data:** Used to teach the model.
- **Test Data:** Used to check how well the model works on new data.

Some projects also use a **Validation Set** to improve the model during training.

Example:
A teacher teaches from one set of questions and gives a different set in the final exam.

5.2 Why is it Important?

The train-test split helps us know if the model works well on new data.

Without a test set, the model might only memorize the training data and give poor results in real life.

Simple Example:
A student who only studies one practice test may not do well on a different exam.

5.3 Common Split Ratios

The most common way to split data is:

Split Ratio	Meaning
80% / 20%	80% for training, 20% for testing (most common).
70% / 30%	More data for testing.
90% / 10%	Used for large datasets.
99% / 1%	Used for very large datasets.

Easy Example:

If you have **1000 records**:

- 800 for training.
- 200 for testing.

This helps make sure the model can work well on new, unseen data.

Key Points to Remember

- **Supervised Learning:** Learns from labeled data.
- **Unsupervised Learning:** Finds patterns in unlabeled data.
- **Overfitting:** Model memorizes data and performs poorly on new data.

- **Prevent Overfitting:** Use simpler models, more data, and proper testing methods.
- **Train-Test Split:** Divide data into training and testing sets to check model performance.
- **80/20 split** is the most commonly used ratio.

6. Case Study: Machine Learning in Healthcare

6.1 Overview

This case study is based on research by **Min, Yu, and Wang (2019)**. The study used Machine Learning to predict whether patients with Chronic Obstructive Pulmonary Disease (COPD) would return to the hospital within 30 days after discharge.

6.2 Problem

Many COPD patients are readmitted to the hospital soon after treatment. Hospitals want to identify high-risk patients early so they can provide extra care and reduce unnecessary readmissions.

Research Question:
Can Machine Learning predict which COPD patients are likely to return to the hospital within 30 days?

6.3 Data and Methods

The researchers collected medical records from **111,992 COPD patients**.

The data included:

- Patient health history.
- Previous hospital visits.
- Medications and other medical information.

They trained several supervised Machine Learning models using historical patient data. The data was divided into training and testing sets to check the accuracy of the models.

Citation: (Min et al., 2019)

6.4 Key Findings

The study found that Machine Learning improved the prediction of hospital readmissions.

The best models combined:

- Medical knowledge from doctors.
- Patterns discovered automatically by Machine Learning.

This helped hospitals identify high-risk patients more accurately.

6.5 Connection to the Data Science Lifecycle

This case study follows the main steps of the Data Science Lifecycle:

1. Define the problem.
2. Collect patient data.
3. Clean and prepare the data.
4. Build Machine Learning models.
5. Test and evaluate the models.

The study mainly focuses on **Model Building** and **Model Evaluation**, showing how supervised learning can solve a real healthcare problem.

Conclusion

Data Science and Machine Learning work together to solve real-world problems. Data Science collects and prepares data, while Machine Learning learns patterns and makes predictions.

The Data Science Lifecycle provides a step-by-step process for building reliable models. Supervised and unsupervised learning are two important learning methods, while techniques such as train-test splitting and overfitting prevention help improve model performance.

The COPD healthcare case study shows how these concepts are used in practice. By analyzing patient data, Machine Learning can predict hospital readmissions and help doctors provide better care. This demonstrates the importance of Data Science and Machine Learning in modern healthcare and many other industries.

Min, X., Yu, B., & Wang, F. (2019). Predictive modeling of the hospital readmission risk from patients' claims data using machine learning: A case study on COPD. Scientific Reports, 9, 2362. https://doi.org/10.1038/s41598-019-39071-y